

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Monday 12 May 2025

Afternoon (Time: 1 hour 40 minutes) **Paper reference** **8FM0/01**

Further Mathematics
Advanced Subsidiary
PAPER 1: Core Pure Mathematics

You must have:
 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Answers without working may not gain full credit.
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Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 11 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
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- Check your answers if you have time at the end.

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Question 1 continued

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(Total for Question 1 is 6 marks)



2.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

$$f(z) = 4z^3 - 12z^2 - 95z + 325$$

Given that $f(-5) = 0$

(a) determine $f(z)$ in the form $(z + a)(bz^2 + cz + d)$ where a, b, c and d are integers. (3)

(b) Hence show that the complex roots of $f(z) = 0$ are $\frac{8 \pm i}{2}$ (2)

(c) Determine the values of z such that $f(2z - 1) = 0$ (2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3.

$$\mathbf{M} = \begin{pmatrix} a & b \\ -1 & -1 \end{pmatrix}$$

where a and b are real constants and $a \neq b$

(a) Determine $\mathbf{M}^{-1}$ in terms of a and b

(3)

Given that $\mathbf{M} + \mathbf{M}^{-1} = \mathbf{I}$, where $\mathbf{I}$ is the 2×2 identity matrix,

(b) determine the value of a and value of b

(3)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 6 marks)



4. (i)

$$\mathbf{P} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

(a) Describe fully the single geometrical transformation P represented by the matrix $\mathbf{P}$.

(2)

(b) State the equation of **one** invariant line under the transformation P .

(1)

(ii)

$$\mathbf{Q} = \begin{pmatrix} \cos 2\theta & 0 \\ 1 & \tan 2\theta \end{pmatrix} \quad \text{where } 0 \leq \theta < 360^\circ$$

The matrix $\mathbf{Q}$ represents the transformation Q .

Triangle T is transformed to triangle T' by the transformation Q .

Given that

- the coordinates of the vertices of T are $(2, 3)$, $(3, 6)$ and $(8, 3)$
- the area of T' is 4.5

determine the possible values of θ

(Solutions relying entirely on calculator technology are not acceptable.)

(7)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

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(Total for Question 4 is 10 marks)



- Given that z satisfies

(a) sketch on an Argand diagram the locus of P as z varies.

(2)

Given that z also satisfies

(b) determine the possible values of θ such that the locus $\arg(z + 1) = \theta$ is a tangent to the locus $|z - 1| = 1$

(3)

- (c) Hence determine the exact possible complex numbers z .

(3)

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 8 marks)



6. Prove by induction that for all positive integers n

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2$$

(6)



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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 6 marks)



7.

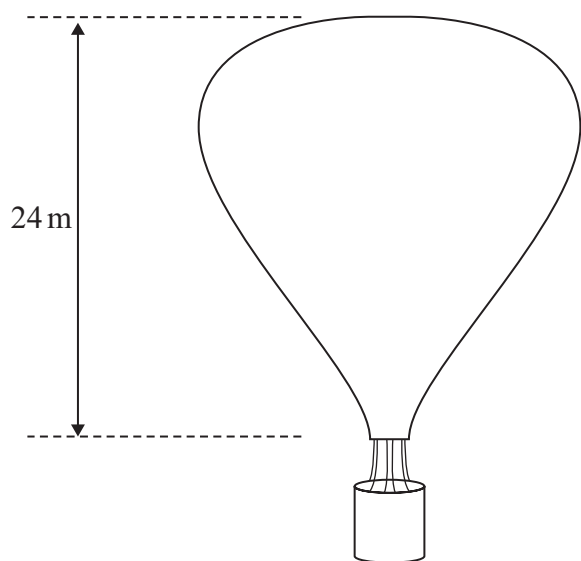


Figure 1

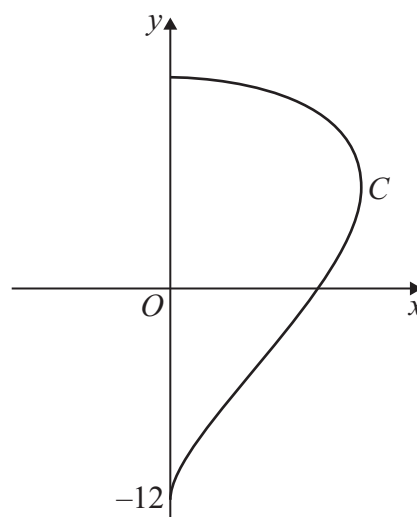


Figure 2

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 1 shows a sketch of a hot air balloon.

When filled with air the balloon has a height of 24 metres.

Figure 2 shows a sketch of the curve C with equation

$$350x^2 = (12 + y)^2 (A - y^2) \quad x \geq 0$$

where A is a constant.

The balloon is modelled by rotating C through 360° about the y -axis.

Given that one y intercept of C is -12

- (a) show that $A = 144$ (1)
- (b) Use algebraic integration to determine the volume of air needed to fill the balloon, according to the model, giving the answer to 2 significant figures. (5)
- (c) Modify the equation $350x^2 = (12 + y)^2 (144 - y^2)$ to model a mathematically similar balloon with a height of 26 metres. (1)
- (d) State one limitation of the models. (1)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8. The first n triangular numbers are

$$1, 3, 6, 10, \dots, \frac{1}{2}n(n+1)$$

where n is a positive integer.

- (a) Use the standard results for $\sum_{r=1}^n r^2$ and $\sum_{r=1}^n r$ to show that the sum of the first n triangular numbers is

$$\frac{1}{6}n(n+1)(n+2) \quad (5)$$

- (b) Hence determine the value of n for which the sum of the first n triangular numbers is $22n$.

(2)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 7 marks)



9. An engineer detects a source of water below the surface of the ground. The engineer models the situation relative to a fixed origin O .

In the model

- the surface of the ground is a plane Π with equation $x - 2y + 8z = 1$
- the source of water is at a point W with coordinates $(6, -2, -4)$

where the units are metres.

- (a) Use the model to determine the shortest distance from W to the surface of the ground. (2)
- (b) By considering the model, comment on whether the answer to part (a) is reliable, giving a reason for your answer. (1)

To access the water, a hole is drilled, in a straight line, from a point P on the surface of the ground to W .

Given that the length of the hole needs to be as short as possible,

- (c) determine the coordinates of P , according to the model. (3)

Given that the actual length of the hole drilled is 2.52 metres,

- (d) use the answer to part (a) to evaluate the model, giving a reason for your answer. (1)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 7 marks)



10. The plane Π_1 has equation $x + y - z = 3$

The plane Π_2 has equation $ax + 3y + 5z = 4$ where a is an integer.

Given that Π_1 is perpendicular to Π_2

(a) determine the value of a .

(2)

The plane Π_3 has equation $x + by + 13z = c$ where b and c are integers.

Given that the three planes form a sheaf,

(b) use algebra to determine the value of b and the value of c .

(6)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Handwriting practice area with horizontal lines.

(Total for Question 10 is 8 marks)



11.

In this question you must show detailed reasoning.

$$f(x) = x^3 - 21x^2 + Ax - 91 \quad \text{where } A \text{ is a real constant}$$

The roots of $f(x) = 0$ are

$$\alpha, \alpha + 3\beta \text{ and } \alpha + 6\beta$$

where α and β are real constants.

Use algebra to determine the value of each of these roots.

(7)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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(Total for Question 11 is 7 marks)

TOTAL FOR CORE PURE MATHEMATICS IS 80 MARKS



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Further Mathematics
Advanced Subsidiary
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1. The cubic equation

$$2x^3 - 3x^2 + 5x + 7 = 0$$

has roots α , β and γ .

Without solving the equation, determine the exact value of

(i) $\alpha^2 + \beta^2 + \gamma^2$ (3)

(ii) $\frac{3}{\alpha} + \frac{3}{\beta} + \frac{3}{\gamma}$ (3)

(iii) $(5 - \alpha)(5 - \beta)(5 - \gamma)$ (3)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 9 marks)



2. $\left[\begin{array}{l} \text{With respect to the **right-hand rule**, a rotation through } \theta^\circ \text{ anticlockwise about} \\ \text{the } z\text{-axis is represented by the matrix} \end{array} \right]$
- $$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Given that the matrix $\mathbf{M}$, where

$$\mathbf{M} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

represents a rotation through α° anticlockwise about the z -axis with respect to the **right-hand rule**,

- (a) determine the value of α . (1)

- (b) Hence determine the smallest possible positive integer value of k for which $\mathbf{M}^k = \mathbf{I}$ (2)

The 3×3 matrix $\mathbf{N}$ represents a reflection in the plane with equation $y = 0$

- (c) Write down the matrix $\mathbf{N}$. (1)

The point A has coordinates $(-2, 4, 3)$

The point B is the image of the point A under the transformation represented by matrix $\mathbf{M}$ followed by the transformation represented by matrix $\mathbf{N}$.

- (d) Show that the coordinates of B are $(2 + \sqrt{3}, 2\sqrt{3} - 1, 3)$ (2)

Given that O is the origin,

- (e) show that, to 3 significant figures, the size of angle AOB is 66.9° (2)

- (f) Hence determine the area of triangle AOB , giving your answer to 3 significant figures. (2)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 10 marks)

3. (a) Use the standard results for summations to show that, for all positive integers n ,

$$\sum_{r=1}^n r^2(r+1) = \frac{1}{12}n(n+1)(n+2)(an+b)$$

where a and b are integers to be determined.

(4)

- (b) Hence show that, for all positive integers k ,

$$\sum_{r=k+1}^{3k} r^2(r+1) = \frac{1}{3}k(3k+1)(Ak^2+Bk+C)$$

where A , B and C are integers to be determined.

(3)

- (c) Hence, using algebra and making your method clear, determine the value of k for which

$$25 \sum_{r=k+1}^{3k} r^2(r+1) = 192k^3(3k+1)$$

(3)

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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 10 marks)

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

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(Total for Question 4 is 8 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 10 marks)



- 6.** The drainage system for a sports field consists of underground pipes.

This situation is modelled with respect to a fixed origin O .

According to the model,

- the surface of the sports field is a plane with equation $z = 0$
- the pipes are straight lines
- one of the pipes, P_1 , passes through the points $A(3, 4, -2)$ and $B(-2, -8, -3)$
- a different pipe, P_2 , has equation $\frac{x-1}{2} = \frac{y-3}{4} = \frac{z+1}{-2}$
- the units are metres

- (a) Determine a vector equation of the line representing the pipe P_1 (2)
- (b) Determine the coordinates of the point at which the pipe P_1 meets the surface of the playing field, according to the model. (2)

Determine, according to the model,

- (c) the acute angle between pipes P_1 and P_2 , giving your answer in degrees to 3 significant figures, (3)
- (d) the shortest distance between pipes P_1 and P_2 (5)

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 12 marks)



7. (i) Prove by induction that, for all positive integers n ,

$$\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1} \quad (5)$$

- (ii) Prove by induction that, for all positive integers n ,

$$f(n) = 3^{2n+4} - 2^{2n}$$

is divisible by 5

(5)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

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(Total for Question 7 is 10 marks)



8.

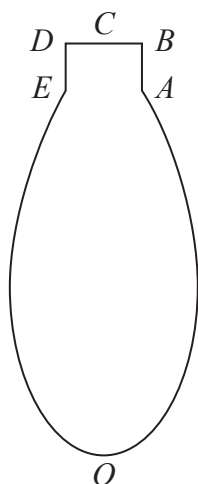


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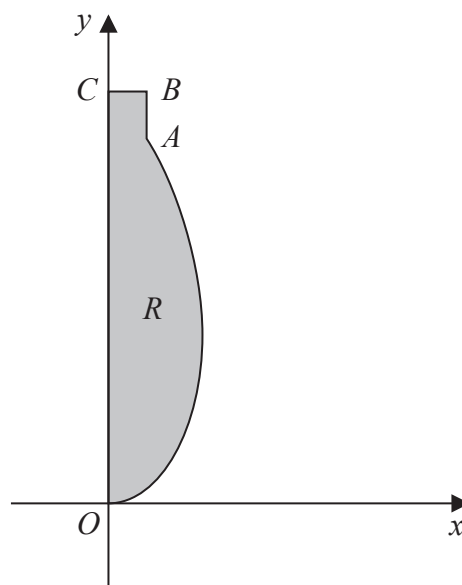


Figure 2

Figure 1 shows the central vertical cross-section, $OABCDEO$, of the design for a solid glass ornament.

Figure 2 shows the finite region, R , which is bounded by the y -axis, the horizontal line CB , the vertical line BA , and the curve AO .

The ornament is formed by rotating the region R through 360° about the y -axis.

The curve AO is modelled by the equation

$$x = ky^2 + \sqrt{y} \quad 0 \leq y \leq 4$$

where k is a constant.

The point A has coordinates $(0.4, 4)$ and the point B has coordinates $(0.4, 4.5)$

The units are centimetres.

(a) Determine the value of k according to this model. (2)

(b) Use algebraic integration to determine the exact volume of glass that would be required to make the ornament, according to the model. (7)

(c) State a limitation of the model. (1)

When the ornament was manufactured, 9 cm^3 of glass was required.

(d) Use this information and your answer to part (b) to evaluate the model, explaining your reasoning. (1)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 11 marks)

TOTAL FOR CORE PURE MATHEMATICS IS 80 MARKS



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Further Mathematics

Advanced Subsidiary

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1.

$$\begin{pmatrix} x & 9 \\ y & z \end{pmatrix} - 3 \begin{pmatrix} z & y \\ z & y \end{pmatrix} = k\mathbf{I}$$

where x , y , z and k are constants.

Determine the value of x , the value of y and the value of z .

(4)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 4 marks)



2. $f(z) = z^3 + az^2 + bz + 175$ where a and b are real constants

Given that $-3 + 4i$ is a root of the equation $f(z) = 0$

(a) determine the value of a and the value of b . (4)

(b) Show all the roots of the equation $f(z) = 0$ on a single Argand diagram. (2)

(c) Write down the roots of the equation $f(z + 2) = 0$ (1)



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Question 2 continued

Lined area for writing the answer to Question 2.

Question 2 continued

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DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3.

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

(a) Describe fully the single geometric transformation A represented by the matrix $\mathbf{A}$. (2)

$$\mathbf{B} = \begin{pmatrix} 1 & 3 & 0 \\ \sqrt{3} & 0 & 5\sqrt{3} \\ 1 & 2 & 0 \end{pmatrix}$$

The transformation B is represented by the matrix $\mathbf{B}$.

The transformation A followed by the transformation B is the transformation C , which is represented by the matrix $\mathbf{C}$.

To determine matrix **C**, a student attempts the following matrix multiplication.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} 1 & 3 & 0 \\ \sqrt{3} & 0 & 5\sqrt{3} \\ 1 & 2 & 0 \end{pmatrix}$$

(b) State the error made by the student. (1)

(c) Determine the correct matrix $\mathbf{C}$. (1)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 4 marks)



4. (i) (a) Show that

$$\frac{2 + 3i}{5 + i} = k(1 + i)$$

where k is a constant to be determined.

(Solutions relying on calculator technology are not acceptable.)

(3)

Given that

- n is a positive integer
- $\left(\frac{2+3i}{5+i}\right)^n$ is a real number

(b) use the answer to part (a) to write down the smallest possible value of n .

(1)

(ii) The complex number $z = a + bi$ where a and b are real constants.

Given that

- $|z^{10}| = 59\,049$
- $\arg(z^{10}) = -\frac{5\pi}{3}$

determine the value of a and the value of b .

(4)

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 8 marks)



5.

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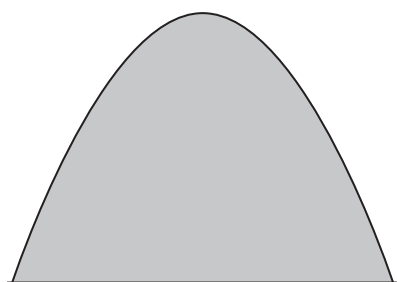


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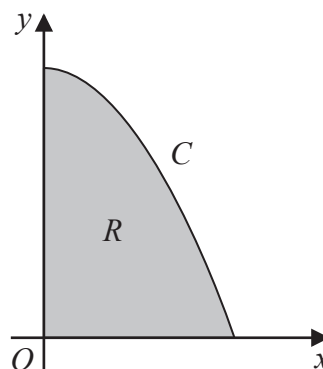


Figure 2

A large pile of concrete waste is created on a building site.

Figure 1 shows a central vertical cross-section of the concrete waste.

The curve C , shown in Figure 2, has equation

$$y + x^2 = 2 \quad 0 \leq x \leq \sqrt{2}$$

The region R , shown shaded in Figure 2, is bounded by the y -axis, the x -axis and the curve C .

The volume of concrete waste is modelled by the volume of revolution formed when R is rotated through 360° about the y -axis. The units are metres.

The density of the concrete waste is 900 kg m^{-3}

- (a) Use the model to estimate the mass of the concrete waste. Give your answer to 2 significant figures.

(6)

- (b) Give a limitation of the model.

(1)

The mass of the concrete waste is approximately 5500 kg.

- (c) Use this information and your answer to part (a) to evaluate the model, giving a reason for your answer.

(1)

DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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DO NOT WRITE IN THIS AREA

Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 8 marks)



6. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$ where λ is a scalar parameter.

The line l_2 is parallel to $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$

(a) Show that l_1 and l_2 are perpendicular.

(2)

The plane Π contains the line l_1 and is perpendicular to $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$

(b) Determine a Cartesian equation of Π

(2)

(c) Verify that the point $A(3, 1, 1)$ lies on Π

(1)

Given that

- the point of intersection of Π and l_2 has coordinates $(2, 3, 2)$
- the point $B(p, q, r)$ lies on l_2
- the distance AB is $2\sqrt{5}$
- p, q and r are positive integers

(d) determine the coordinates of B .

(6)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 11 marks)



7. (i) Shade, on an Argand diagram, the set of points for which

$$|z - 3| \leq |z + 6i|$$

(3)

- (ii) Determine the exact complex number w which satisfies both

$$\arg(w - 2) = \frac{\pi}{3} \quad \text{and} \quad \arg(w + 1) = \frac{\pi}{6}$$

(6)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 9 marks)



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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 8 marks)



9. (i)
$$\mathbf{P} = \begin{pmatrix} k & -2 & 7 \\ -3 & -5 & 2 \\ k & k & 4 \end{pmatrix} \quad \text{where } k \text{ is a constant}$$

Show that $\mathbf{P}$ is non-singular for all real values of k .

(4)

(ii)
$$\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ -3 & 0 \end{pmatrix}$$

The matrix $\mathbf{Q}$ represents a linear transformation T

Under T , the point $A(a, 2)$ and the point $B(4, -a)$, where a is a constant, are transformed to the points A' and B' respectively.

Given that the distance $A'B'$ is $\sqrt{58}$, determine the possible values of a .

(5)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 9 marks)



10.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i) The quartic equation

$$z^4 + 5z^2 - 30 = 0$$

has roots p, q, r and s .

Without solving the equation, determine the quartic equation whose roots are

$$(3p - 1), (3q - 1), (3r - 1) \text{ and } (3s - 1)$$

Give your answer in the form $w^4 + aw^3 + bw^2 + cw + d = 0$, where a, b, c and d are integers to be found.

(5)

(ii) The roots of the cubic equation

$$4x^3 + nx + 81 = 0 \quad \text{where } n \text{ is a real constant}$$

are $\alpha, 2\alpha$ and $\alpha - \beta$

Determine

(a) the values of the roots of the equation,

(5)

(b) the value of n .

(2)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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(Total for Question 10 is 12 marks)

TOTAL FOR PAPER IS 80 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Time 1 hour 40 minutes

Paper
reference

8FM0/01

Further Mathematics

Advanced Subsidiary

PAPER 1: Core Pure Mathematics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Q:1/1/1/1/



Pearson

1.

$$\mathbf{A} = \begin{pmatrix} 4 & -1 \\ 7 & 2 \\ -5 & 8 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 2 & 3 & 2 \\ -1 & 6 & 5 \end{pmatrix} \quad \mathbf{C} = \begin{pmatrix} -5 & 2 & 1 \\ 4 & 3 & 8 \\ -6 & 11 & 2 \end{pmatrix}$$

Given that $\mathbf{I}$ is the 3×3 identity matrix,

- (a) (i) show that there is an integer k for which

$$\mathbf{AB} - 3\mathbf{C} + k\mathbf{I} = \mathbf{0}$$

stating the value of k

- (ii) explain why there can be no constant m such that

$$\mathbf{BA} - 3\mathbf{C} + m\mathbf{I} = \mathbf{0}$$

(4)

- (b) (i) Show how the matrix $\mathbf{C}$ can be used to solve the simultaneous equations

$$-5x + 2y + z = -14$$

$$4x + 3y + 8z = 3$$

$$-6x + 11y + 2z = 7$$

- (ii) Hence use your calculator to solve these equations.

(3)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 7 marks)



2. (a) Express the complex number $w = 4\sqrt{3} - 4i$ in the form $r(\cos \theta + i \sin \theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$ (4)
- (b) Show, on a single Argand diagram,
- (i) the point representing w
- (ii) the locus of points defined by $\arg(z + 10i) = \frac{\pi}{3}$ (3)
- (c) Hence determine the minimum distance of w from the locus $\arg(z + 10i) = \frac{\pi}{3}$ (3)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 10 marks)



3. With respect to the **right-hand rule**, a rotation through θ° anticlockwise about the y -axis is represented by the matrix
- $$\begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

The point P has coordinates $(8, 3, 2)$

The point Q is the image of P under the transformation reflection in the plane $y = 0$

- (a) Write down the coordinates of Q (1)

The point R is the image of P under the transformation rotation through 120° anticlockwise about the y -axis, with respect to the **right-hand rule**.

- (b) Determine the exact coordinates of R (2)

- (c) Hence find $|\vec{PR}|$ giving your answer as a simplified surd. (2)

- (d) Show that $\overrightarrow{PR}$ and $\overrightarrow{PQ}$ are perpendicular. (1)

- (e) Hence determine the exact area of triangle PQR , giving your answer as a surd in simplest form.



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 8 marks)



4. The roots of the quartic equation

$$3x^4 + 5x^3 - 7x + 6 = 0$$

are α , β , γ and δ

Making your method clear and without solving the equation, determine the exact value of

(i) $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ (3)

(ii) $\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma} + \frac{2}{\delta}$ (3)

(iii) $(3 - \alpha)(3 - \beta)(3 - \gamma)(3 - \delta)$ (3)



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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 9 marks)



5. (a) Use the standard summation formulae to show that, for $n \in \mathbb{N}$,

$$\sum_{r=1}^n (3r^2 - 17r - 25) = n(n^2 - An - B)$$

where A and B are integers to be determined.

(4)

- (b) Explain why, for $k \in \mathbb{N}$,

$$\sum_{r=1}^{3k} r \tan(60r)^\circ = -k\sqrt{3}$$

(2)

Using the results from part (a) and part (b) and showing all your working,

- (c) determine any value of n that satisfies

$$\sum_{r=5}^n (3r^2 - 17r - 25) = 15 \left[\sum_{r=6}^{3n} r \tan(60r)^\circ \right]^2$$

(6)



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Question 5 continued

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DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 12 marks)



6. The surface of a horizontal tennis court is modelled as part of a horizontal plane, with the origin on the ground at the centre of the court, and

- $\mathbf{i}$ and $\mathbf{j}$ are unit vectors directed across the width and length of the court respectively
- $\mathbf{k}$ is a unit vector directed vertically upwards
- units are metres

After being hit, a tennis ball, modelled as a particle, moves along the path with equation

$$\mathbf{r} = (-4.1 + 9\lambda - 2.3\lambda^2)\mathbf{i} + (-10.25 + 15\lambda)\mathbf{j} + (0.84 + 0.8\lambda - \lambda^2)\mathbf{k}$$

where λ is a scalar parameter with $\lambda \geq 0$

Assuming that the tennis ball continues on this path until it hits the ground,

- (a) find the value of λ at the point where the ball hits the ground. (2)

The direction in which the tennis ball is moving at a general point on its path is given by

$$(9 - 4.6\lambda)\mathbf{i} + 15\mathbf{j} + (0.8 - 2\lambda)\mathbf{k}$$

- (b) Write down the direction in which the tennis ball is moving as it hits the ground. (1)
- (c) Hence find the acute angle at which the tennis ball hits the ground, giving your answer in degrees to one decimal place. (4)

The net of the tennis court lies in the plane $\mathbf{r} \cdot \mathbf{j} = 0$

- (d) Find the position of the tennis ball at the point where it is in the same plane as the net. (3)

The maximum height above the court of the top of the net is 0.9 m.

Modelling the top of the net as a horizontal straight line,

- (e) state whether the tennis ball will pass over the net according to the model, giving a reason for your answer. (1)

With reference to the model,

- (f) decide whether the tennis ball will actually pass over the net, giving a reason for your answer. (2)



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Question 6 continued

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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 13 marks)



7. Prove by mathematical induction that, for $n \in \mathbb{N}$

$$\begin{pmatrix} -5 & 9 \\ -4 & 7 \end{pmatrix}^n = \begin{pmatrix} 1 - 6n & 9n \\ -4n & 1 + 6n \end{pmatrix}$$

(6)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 6 marks)



8.

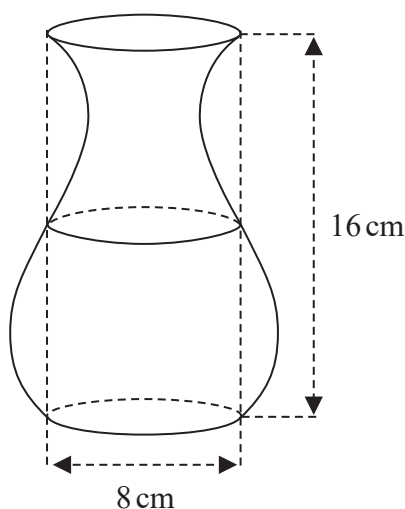


Figure 1

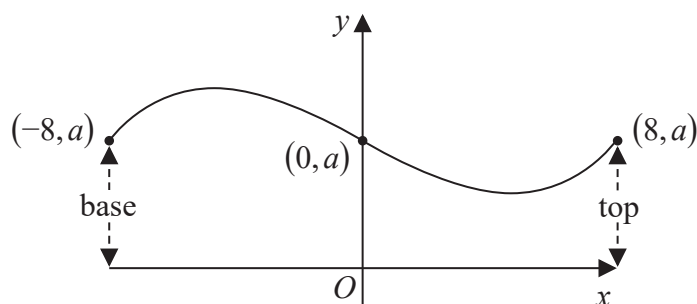


Figure 2

Figure 1 shows a sketch of a 16 cm tall vase which has a flat circular base with diameter 8 cm and a circular opening of diameter 8 cm at the top.

A student measures the circular cross-section halfway up the vase to be 8 cm in diameter.

The student models the shape of the vase by rotating a curve, shown in Figure 2, through 360° about the x -axis.

(a) State the value of a that should be used when setting up the model.

(1)

Two possible equations are suggested for the curve in the model.

$$\text{Model A} \quad y = a - 2 \sin\left(\frac{45}{2}x\right)$$

$$\text{Model B} \quad y = a + \frac{x(x-8)(x+8)}{100}$$

For each model,

- (b) (i) find the distance from the base at which the widest part of the vase occurs,
 (ii) find the diameter of the vase at this widest point.

(7)

The widest part of the vase has diameter 12 cm and is just over 3 cm from the base.

- (c) Using this information and making your reasoning clear, suggest which model is more appropriate.

(1)

- (d) Using algebraic integration, find the volume for the vase predicted by Model B. You must make your method clear.

(5)

The student pours water from a full one litre jug into the vase and finds that there is 100 ml left in the jug when the vase is full.

- (e) Comment on the suitability of Model B in light of this information.

(1)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

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(Total for Question 8 is 15 marks)

TOTAL FOR CORE PURE MATHEMATICS IS 80 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Time 1 hour 40 minutes	Paper reference	8FM0/01	
Further Mathematics Advanced Subsidiary PAPER 1: Core Pure Mathematics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►



1. $\mathbf{P} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ $\mathbf{Q} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$

(a) (i) Describe fully the single geometrical transformation P represented by the matrix $\mathbf{P}$.

(ii) Describe fully the single geometrical transformation Q represented by the matrix $\mathbf{Q}$. (4)

The transformation P followed by the transformation Q is the transformation R , which is represented by the matrix $\mathbf{R}$.

(b) Determine $\mathbf{R}$. (1)

(c) (i) Evaluate the determinant of $\mathbf{R}$.

(ii) Explain how the value obtained in (c)(i) relates to the transformation R . (2)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2. The cubic equation

$$9x^3 - 5x^2 + 4x + 7 = 0$$

has roots α , β and γ .

Without solving the equation, find the cubic equation whose roots are $(3\alpha - 2)$, $(3\beta - 2)$ and $(3\gamma - 2)$, giving your answer in the form $aw^3 + bw^2 + cw + d = 0$, where a , b , c and d are integers to be determined.

(5)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. (a) Use the standard results for summations to show that for all positive integers n

$$\sum_{r=1}^n (5r-2)^2 = \frac{1}{6}n(an^2 + bn + c)$$

where a , b and c are integers to be determined.

(5)

- (b) Hence determine the value of k for which

$$\sum_{r=1}^k (5r-2)^2 = 94k^2$$

(4)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4.

$$\mathbf{M} = \begin{pmatrix} 2 & 1 & 4 \\ k & 2 & -2 \\ 4 & 1 & -2 \end{pmatrix}$$

$$\mathbf{N} = \begin{pmatrix} k-7 & 6 & -10 \\ 2 & -20 & 24 \\ -3 & 2 & -1 \end{pmatrix}$$

where k is a constant.

(a) Determine, in simplest form in terms of k , the matrix $\mathbf{MN}$. (2)

(b) Given that $k = 5$

(i) write down $\mathbf{MN}$

(ii) hence write down $\mathbf{M}^{-1}$ (2)

(c) Solve the simultaneous equations

$$2x + y + 4z = 2$$

$$5x + 2y - 2z = 3$$

$$4x + y - 2z = -1$$
(2)

(d) Interpret the answer to part (c) geometrically. (1)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 7 marks)



5.

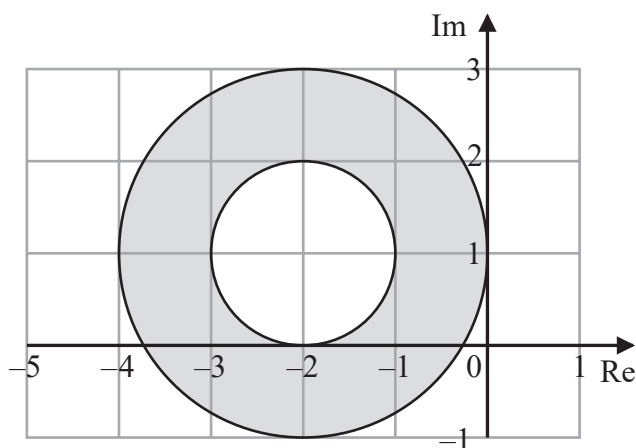


Figure 1

Figure 1 shows an Argand diagram.

The set P , of points that lie within the shaded region including its boundaries, is defined by

$$P = \{z \in \mathbb{C} : a \leq |z + b + ci| \leq d\}$$

where a , b , c and d are integers.

(a) Write down the values of a , b , c and d .

(3)

The set Q is defined by

$$Q = \{z \in \mathbb{C} : a \leq |z + b + ci| \leq d\} \cap \{z \in \mathbb{C} : |z - i| \leq |z - 3i|\}$$

(b) Determine the exact area of the region defined by Q , giving your answer in simplest form.

(7)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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6. A mining company has identified a mineral layer below ground.

The mining company wishes to drill down to reach the mineral layer and models the situation as follows.

With respect to a fixed origin O ,

- the ground is modelled as a horizontal plane with equation $z = 0$
- the mineral layer is modelled as part of the plane containing the points $A(10, 5, -50)$, $B(15, 30, -45)$ and $C(-5, 20, -60)$, where the units are in metres

- (a) Determine an equation for the plane containing A , B and C , giving your answer in the form $\mathbf{r} \cdot \mathbf{n} = d$ (5)
- (b) Determine, according to the model, the acute angle between the ground and the plane containing the mineral layer. Give your answer to the nearest degree. (3)

The mining company plans to drill vertically downwards from the point $(5, 12, 0)$ on the ground to reach the mineral layer.

- (c) Using the model, determine, in metres to 1 decimal place, the distance the mining company will need to drill in order to reach the mineral layer. (2)
- (d) State a limitation of the assumption that the mineral layer can be modelled as a plane. (1)



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Question 6 continued

Handwriting practice area with 30 horizontal lines.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 11 marks)



7.

$$f(z) = z^4 - 6z^3 + pz^2 + qz + r$$

where p , q and r are real constants.

The roots of the equation $f(z) = 0$ are α , β , γ and δ where $\alpha = 3$ and $\beta = 2 + i$

Given that γ is a complex root of $f(z) = 0$

(a) (i) write down the root γ ,

(ii) explain why δ must be real.

(2)

(b) Determine the value of δ .

(2)

(c) Hence determine the values of p , q and r .

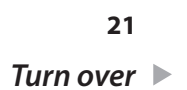
(3)

(d) Write down the roots of the equation $f(-2z) = 0$

(2)



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Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 9 marks)



8. (a) Prove by induction that, for all positive integers n ,

$$\sum_{r=1}^n r(r+1)(2r+1) = \frac{1}{2} n(n+1)^2(n+2) \quad (6)$$

- (b) Hence, show that, for all positive integers n ,

$$\sum_{r=n}^{2n} r(r+1)(2r+1) = \frac{1}{2} n(n+1)(an+b)(cn+d)$$

where a , b , c and d are integers to be determined.

(3)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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9.

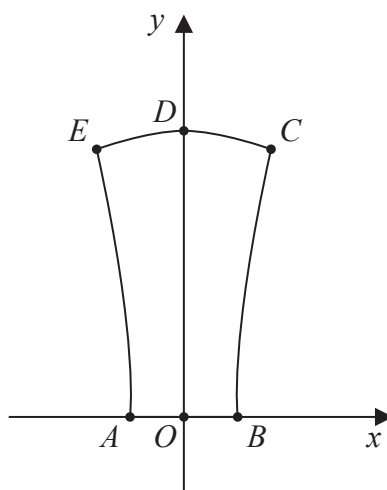


Figure 2

Figure 2 shows the vertical cross-section, $AOBCE$, through the centre of a wax candle.

In a model, the candle is formed by rotating the region bounded by the y -axis, the line OB , the curve BC , and the curve CD through 360° about the y -axis.

The point B has coordinates $(3, 0)$ and the point C has coordinates $(5, 15)$.

The units are in centimetres.

The curve BC is represented by the equation

$$y = \frac{\sqrt{225x^2 - 2025}}{a} \quad 3 \leq x < 5$$

where a is a constant.

(a) Determine the value of a according to this model.

(2)

The curve CD is represented by the equation

$$y = 16 - 0.04x^2 \quad 0 \leq x < 5$$

(b) Using algebraic integration, determine, according to the model, the exact volume of wax that would be required to make the candle.

(9)

(c) State a limitation of the model.

(1)

When the candle was manufactured, 700 cm^3 of wax were required.

(d) Use this information and your answer to part (b) to evaluate the model, explaining your reasoning.

(1)



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Question 9 continued

Handwriting practice area with horizontal lines.



Question 9 continued

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Question 9 continued

Handwriting practice area with horizontal lines.



Question 9 continued

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(Total for Question 9 is 13 marks)

TOTAL FOR CORE PURE MATHEMATICS IS 80 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Monday 5 Oct 2020

Afternoon (Time: 1 hour 40 minutes)

Paper Reference **8FM0/01**

Further Mathematics

Advanced Subsidiary

Paper 1: Core Pure Mathematics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Information

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- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

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1. A system of three equations is defined by

$$\begin{aligned} kx + 3y - z &= 3 \\ 3x - y + z &= -k \\ -16x - ky - kz &= k \end{aligned}$$

where k is a positive constant.

Given that there is no unique solution to all three equations,

- (a) show that $k = 2$

(2)

Using $k = 2$

- (b) determine whether the three equations are consistent, justifying your answer.

(3)

- (c) Interpret the answer to part (b) geometrically.

(1)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. Given that

$$\begin{aligned} z_1 &= 2 + 3i \\ |z_1 z_2| &= 39\sqrt{2} \\ \arg(z_1 z_2) &= \frac{\pi}{4} \end{aligned}$$

where z_1 and z_2 are complex numbers,

(a) write z_1 in the form $r(\cos \theta + i \sin \theta)$

Give the exact value of r and give the value of θ in radians to 4 significant figures.

(2)

(b) Find z_7 , giving your answer in the form $a + ib$ where a and b are integers.

(6)

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



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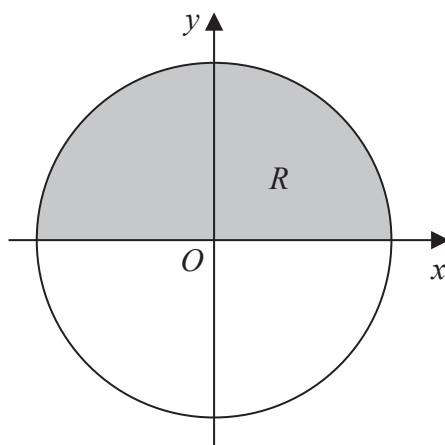


Figure 1

Figure 1 shows a circle with radius r and centre at the origin.

The region R , shown shaded in Figure 1, is bounded by the x -axis and the part of the circle for which $y > 0$

The region R is rotated through 360° about the x -axis to create a sphere with volume V

Use integration to show that $V = \frac{4}{3}\pi r^3$

(5)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



4.

All units in this question are in metres.

A lawn is modelled as a plane that contains the points $L(-2, -3, -1)$, $M(6, -2, 0)$ and $N(2, 0, 0)$, relative to a fixed origin O .

- (a) Determine a vector equation of the plane that models the lawn, giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$ (3)

- (b) (i) Show that, according to the model, the lawn is perpendicular to the vector $\begin{pmatrix} 1 \\ 2 \\ -10 \end{pmatrix}$
- (ii) Hence determine a Cartesian equation of the plane that models the lawn.
- (4)**

There are two posts set in the lawn.

There is a washing line between the two posts.

The washing line is modelled as a straight line through points at the top of each post with coordinates $P(-10, 8, 2)$ and $Q(6, 4, 3)$.

- (c) Determine a vector equation of the line that models the washing line. **(2)**
- (d) State a limitation of one of the models. **(1)**

The point $R(2, 5, 2.75)$ lies on the washing line.

- (e) Determine, according to the model, the shortest distance from the point R to the lawn, giving your answer to the nearest cm. (2)

Given that the shortest distance from the point R to the lawn is actually 1.5 m,

- (f) use your answer to part (e) to evaluate the model, explaining your reasoning. (1)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 13 marks)



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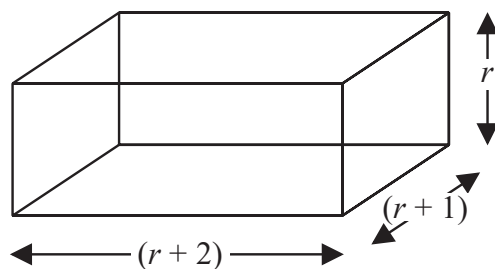


Figure 2

A block has length $(r + 2)$ cm, width $(r + 1)$ cm and height r cm, as shown in Figure 2.

In a set of n such blocks, the first block has a height of 1 cm, the second block has a height of 2 cm, the third block has a height of 3 cm and so on.

- (a) Use the standard results for $\sum_{r=1}^n r^3$, $\sum_{r=1}^n r^2$ and $\sum_{r=1}^n r$ to show that the **total** volume, V , of all n blocks in the set is given by

$$V = \frac{n}{4}(n+1)(n+2)(n+3) \quad n \geq 1 \quad (5)$$

Given that the total volume of all n blocks is

$$(n^4 + 6n^3 - 11\,710) \text{ cm}^3$$

- (b) determine how many blocks make up the set. (2)

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Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 7 marks)



6. (i)

$$\mathbf{A} = \begin{pmatrix} 2 & a \\ a-4 & b \end{pmatrix}$$

where a and b are non-zero constants.

Given that the matrix $\mathbf{A}$ is self-inverse,

- (a) determine the value of b and the possible values for a . (5)

The matrix $\mathbf{A}$ represents a linear transformation M .

Using the smaller value of a from part (a),

- (b) show that the invariant points of the linear transformation M form a line, stating the equation of this line. (3)

(ii)

$$\mathbf{P} = \begin{pmatrix} p & 2p \\ -1 & 3p \end{pmatrix}$$

where p is a positive constant.

The matrix $\mathbf{P}$ represents a linear transformation U .

The triangle T has vertices at the points with coordinates $(1, 2)$, $(3, 2)$ and $(2, 5)$.

The area of the image of T under the linear transformation U is 15

- (a) Determine the value of p . (4)

The transformation V consists of a stretch scale factor 3 parallel to the x -axis with the y -axis invariant followed by a stretch scale factor -2 parallel to the y -axis with the x -axis invariant. The transformation V is represented by the matrix $\mathbf{Q}$.

- (b) Write down the matrix $\mathbf{Q}$. (2)

Given that U followed by V is the transformation W , which is represented by the matrix $\mathbf{R}$,

- (c) find the matrix $\mathbf{R}$. (2)



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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 16 marks)



7.

$$f(z) = z^4 + az^3 + bz^2 + cz + d$$

where a , b , c and d are real constants.

The equation $f(z) = 0$ has complex roots z_1 , z_2 , z_3 and z_4

When plotted on an Argand diagram, the points representing z_1 , z_2 , z_3 and z_4 form the vertices of a square, with one vertex in each quadrant.

Given that $z_1 = 2 + 3i$, determine the values of a , b , c and d .

(6)



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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 6 marks)



8. Prove by induction that, for $n \in \mathbb{Z}^+$

$$f(n) = 2^{n+2} + 3^{2n+1}$$

is divisible by 7

(6)



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9. The cubic equation

$$3x^3 + x^2 - 4x + 1 = 0$$

has roots α , β , and γ .

Without solving the cubic equation,

(a) determine the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ (3)

(b) find a cubic equation that has roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$, giving your answer in the form

$x^3 + ax^2 + bx + c = 0$, where a , b and c are integers to be determined. (3)



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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 6 marks)



10. Given that there are two distinct complex numbers z that satisfy

$$\{z: |z - 3 - 5i| = 2r\} \cap \left\{z: \arg(z - 2) = \frac{3\pi}{4}\right\}$$

determine the exact range of values for the real constant r .

(7)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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(Total for Question 10 is 7 marks)

TOTAL FOR CORE PURE MATHEMATICS IS 80 MARKS



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Monday 13 May 2019			
Afternoon (Time: 1 hour 40 minutes)		Paper Reference 8FM0-01	
Further Mathematics Advanced Subsidiary Paper 1: Core Pure Mathematics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



2. The cubic equation

$$2x^3 + 6x^2 - 3x + 12 = 0$$

has roots α , β and γ .

Without solving the equation, find the cubic equation whose roots are $(\alpha + 3)$, $(\beta + 3)$ and $(\gamma + 3)$, giving your answer in the form $pw^3 + qw^2 + rw + s = 0$, where p , q , r and s are integers to be found.

(5)



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3. Prove by mathematical induction that, for $n \in \mathbb{N}$

$$\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1}$$

(6)



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4. The line l has equation

$$\frac{x+2}{1} = \frac{y-5}{-1} = \frac{z-4}{-3}$$

The plane Π has equation

$$\mathbf{r} \cdot (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = -7$$

Determine whether the line l intersects II at a single point, or lies in II , or is parallel to II without intersecting it.

(5)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 5 marks)



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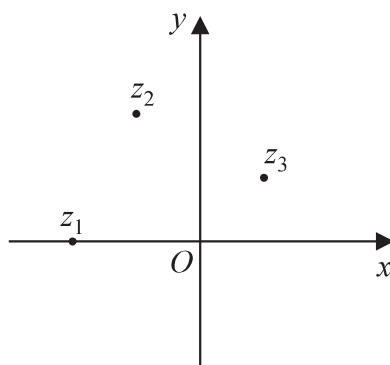


Figure 1

The complex numbers $z_1 = -2$, $z_2 = -1 + 2i$ and $z_3 = 1 + i$ are plotted in Figure 1, on an Argand diagram for the complex plane with $z = x + iy$

- (a) Explain why z_1 , z_2 and z_3 cannot all be roots of a quartic polynomial equation with real coefficients.

(2)

- (b) Show that $\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \frac{\pi}{4}$

(3)

- (c) Hence show that $\arctan(2) - \arctan\left(\frac{1}{3}\right) = \frac{\pi}{4}$

(2)

A copy of Figure 1, labelled Diagram 1, is given on page 12.

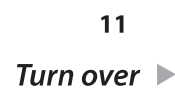
- (d) Shade, on Diagram 1, the set of points of the complex plane that satisfy the inequality

$$|z + 2| \leq |z - 1 - i|$$

(2)



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Question 5 continued

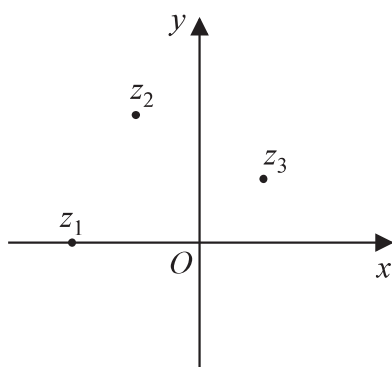


Diagram 1

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[illegible]

- 6.** An art display consists of an arrangement of n marbles.

When arranged in ascending order of mass, the mass of the first marble is 10 grams. The mass of each subsequent marble is 3 grams more than the mass of the previous one, so that the r th marble has mass $(7 + 3r)$ grams.

- (a) Show that the mean mass, in grams, of the marbles in the display is given by

$$\frac{1}{2}(3n+17) \tag{3}$$

Given that there are 85 marbles in the display,

- (b) use the standard summation formulae to find the standard deviation of the mass of the marbles in the display, giving your answer, in grams, to one decimal place. (6)

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 9 marks)



7. $f(z) = z^3 - 8z^2 + pz - 24$

where p is a real constant.

Given that the equation $f(z) = 0$ has distinct roots

$$\alpha, \beta \text{ and } \left(\alpha + \frac{12}{\alpha} - \beta \right)$$

(a) solve completely the equation $f(z) = 0$ (6)

(b) Hence find the value of p . (2)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8. A gas company maintains a straight pipeline that passes under a mountain.

The pipeline is modelled as a straight line and one side of the mountain is modelled as a plane.

There are accessways from a control centre to two access points on the pipeline.

Modelling the control centre as the origin O , the two access points on the pipeline have coordinates $P(-300, 400, -150)$ and $Q(300, 300, -50)$, where the units are metres.

- (a) Find a vector equation for the line PQ , giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, where λ is a scalar parameter.

(2)

The equation of the plane modelling the side of the mountain is $2x + 3y - 5z = 300$

The company wants to create a new accessway from this side of the mountain to the pipeline.

The accessway will consist of a tunnel of shortest possible length between the pipeline and the point $M(100, k, 100)$ on this side of the mountain, where k is a constant.

- (b) Using the model, find

- (i) the coordinates of the point at which this tunnel will meet the pipeline,
- (ii) the length of this tunnel.

(7)

It is only practical to construct the new accessway if it will be significantly shorter than both of the existing accessways, OP and OQ .

- (c) Determine whether the company should build the new accessway.

(2)

- (d) Suggest one limitation of the model.

(1)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 12 marks)



9. $f(x) = 2x^{\frac{1}{3}} + x^{-\frac{2}{3}} \quad x > 0$

The finite region bounded by the curve $y = f(x)$, the line $x = \frac{1}{8}$, the x -axis and the line $x = 8$ is rotated through θ radians about the x -axis to form a solid of revolution.

Given that the volume of the solid formed is $\frac{461}{2}$ units cubed, use algebraic integration to find the angle θ through which the region is rotated.

(8)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

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(Total for Question 9 is 8 marks)



10. The population of chimpanzees in a particular country consists of juveniles and adults. Juvenile chimpanzees do not reproduce.

In a study, the numbers of juvenile and adult chimpanzees were estimated at the start of each year. A model for the population satisfies the matrix system

$$\begin{pmatrix} J_{n+1} \\ A_{n+1} \end{pmatrix} = \begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix} \begin{pmatrix} J_n \\ A_n \end{pmatrix} \quad n = 0, 1, 2, \dots$$

where a is a constant, and J_n and A_n are the respective numbers of juvenile and adult chimpanzees n years after the start of the study.

- (a) Interpret the meaning of the constant a in the context of the model.

(1)

At the start of the study, the total number of chimpanzees in the country was estimated to be 64 000

According to the model, after one year the number of juvenile chimpanzees is 15 360 and the number of adult chimpanzees is 43 008

- (b) (i) Find, in terms of a

$$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1}$$

(3)

- (ii) Hence, or otherwise, find the value of a .

(3)

- (iii) Calculate the change in the number of juvenile chimpanzees in the first year of the study, according to this model.

(2)

Given that the number of juvenile chimpanzees is known to be in decline in the country,

- (c) comment on the short-term suitability of this model.

(1)

A study of the population revealed that adult chimpanzees stop reproducing at the age of 40 years.

- (d) Refine the matrix system for the model to reflect this information, giving a reason for your answer.

(There is no need to estimate any unknown values for the refined model, but any known values should be made clear.)

(2)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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(Total for Question 10 is 12 marks)

TOTAL FOR CORE PURE MATHEMATICS IS 80 MARKS



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Further Mathematics

Advanced Subsidiary

Paper 1: Core Pure Mathematics

Monday 14 May 2018 – Afternoon

Time: 1 hour 40 minutes

Paper Reference

8FM0/01

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

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(a) Find $\mathbf{M}^{-1}$ giving each element in exact form.

(2)

$$2x + y - 3z = -4$$

$$4x - 2y + z = 9$$

$$3x + 5y - 2z = 5$$

(2)

(1)

Question 1 continued

Lined area for writing the answer to Question 1 continued.

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Question 1 continued

Lined area for writing the answer to Question 1.

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2. The cubic equation

$$z^3 - 3z^2 + z + 5 = 0$$

has roots α , β and γ .

Without solving the equation, find the cubic equation whose roots are $(2\alpha + 1)$, $(2\beta + 1)$ and $(2\gamma + 1)$, giving your answer in the form $w^3 + pw^2 + qw + r = 0$, where p , q and r are integers to be found.

(5)

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Question 2 continued

(Total for Question 2 is 5 marks)



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3. (a) Shade on an Argand diagram the set of points

$$\{z \in \mathbb{C} : |z - 1 - i| \leq 3\} \cap \left\{z \in \mathbb{C} : \frac{\pi}{4} \leq \arg(z - 2) \leq \frac{3\pi}{4}\right\} \quad (5)$$

The complex number w satisfies

$$|w - 1 - i| = 3 \text{ and } \arg(w - 2) = \frac{\pi}{4}$$

(b) Find, in simplest form, the exact value of $|w|^2$

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Question 3 continued

Lined area for writing the answer to Question 3.

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Question 3 continued

Lined area for writing the answer to Question 3.

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 9 marks)



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4. Part of the mains water system for a housing estate consists of water pipes buried beneath the ground surface. The water pipes are modelled as straight line segments. One water pipe, W , is buried beneath a particular road. With respect to a fixed origin O , the road surface is modelled as a plane with equation $3x - 5y - 18z = 7$, and W passes through the points $A(-1, -1, -3)$ and $B(1, 2, -3)$. The units are in metres.

(a) Use the model to calculate the acute angle between W and the road surface.

(5)

A point $C(-1, -2, 0)$ lies on the road. A section of water pipe needs to be connected to W from C .

(b) Using the model, find, to the nearest cm, the shortest length of pipe needed to connect C to W .

(6)



Question 4 continued

Lined area for writing the answer to Question 4 continued.

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Question 4 continued

Lined area for writing the answer to Question 4.

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Question 4 continued

Handwriting practice area with 25 horizontal lines.

(Total for Question 4 is 11 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.

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Question 5 continued

Lined area for writing the answer to Question 5.

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Question 5 continued

Handwriting practice area with 30 horizontal lines.

(Total for Question 5 is 10 marks)



6. (a) Use the standard results for $\sum_{r=1}^n r^2$ and $\sum_{r=1}^n r$ to show that

$$\sum_{r=1}^n (3r-2)^2 = \frac{1}{2}n[6n^2 - 3n - 1]$$

for all positive integers n .

(5)

(b) Hence find any values of n for which

$$\sum_{r=5}^n (3r-2)^2 + 103 \sum_{r=1}^{28} r \cos\left(\frac{r\pi}{2}\right) = 3n^3$$

(5)

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Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 10 marks)



7.

$$f(z) = z^3 + z^2 + pz + q$$

where p and q are real constants.

The equation $f(z) = 0$ has roots z_1, z_2 and z_3

When plotted on an Argand diagram, the points representing z_1, z_2 and z_3 form the vertices of a triangle of area 35

Given that $z_1 = 3$, find the values of p and q .

(7)

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Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

Handwriting practice area with 25 horizontal lines.

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Question 7 continued

Handwriting practice area with 28 horizontal lines.

(Total for Question 7 is 7 marks)



8. (i) Prove by induction that for $n \in \mathbb{Z}^+$

$$\begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}^n = \begin{pmatrix} 4n+1 & -8n \\ 2n & 1-4n \end{pmatrix} \quad (6)$$

(ii) Prove by induction that for $n \in \mathbb{Z}^+$

$$f(n) = 4^{n+1} + 5^{2n-1}$$

is divisible by 21

(6)

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Question 8 continued

Handwriting practice area with 20 horizontal lines.

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Question 8 continued

Lined area for writing the answer to Question 8.

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Question 8 continued

Handwriting practice area with 24 horizontal lines.

(Total for Question 8 is 12 marks)



9.

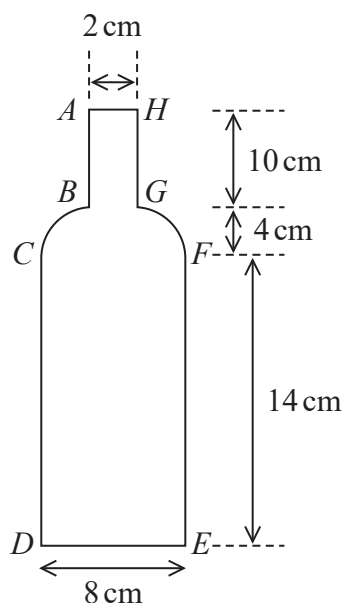


Figure 1

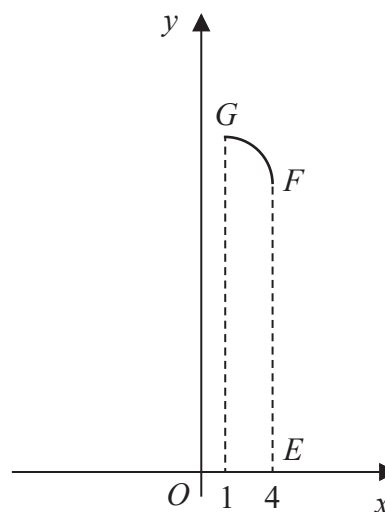


Figure 2

A mathematics student is modelling the profile of a glass bottle of water. Figure 1 shows a sketch of a central vertical cross-section $ABCDEFGHA$ of the bottle with the measurements taken by the student.

The horizontal cross-section between CF and DE is a circle of diameter 8 cm and the horizontal cross-section between BG and AH is a circle of diameter 2 cm.

The student thinks that the curve GF could be modelled as a curve with equation

$$y = ax^2 + b \quad 1 \leq x \leq 4$$

where a and b are constants and O is the fixed origin, as shown in Figure 2.

(a) Find the value of a and the value of b according to the model.

(2)

(b) Use the model to find the volume of water that the bottle can contain.

(7)

(c) State a limitation of the model.

(1)

The label on the bottle states that the bottle holds approximately 750 cm^3 of water.

(d) Use this information and your answer to part (b) to evaluate the model, explaining your reasoning.

(1)



Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Lined area for writing answers.

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(Total for Question 9 is 11 marks)

TOTAL FOR PAPER IS 80 MARKS



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Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Further Mathematics

Advanced Subsidiary

Paper 1: Core Pure Mathematics

Sample Assessment Material for first teaching September 2017

Time: 1 hour 40 minutes

Paper Reference

8FM0/01

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

Answer ALL questions. Write your answers in the spaces provided.

1. $f(z) = z^3 + pz^2 + qz - 15$

where p and q are real constants.

Given that the equation $f(z) = 0$ has roots

$$\alpha, \frac{5}{\alpha} \text{ and } \left(\alpha + \frac{5}{\alpha} - 1\right)$$

(a) solve completely the equation $f(z) = 0$

(5)

(b) Hence find the value of p .

(2)

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[illegible]

2. The plane Π passes through the point A and is perpendicular to the vector $\mathbf{n}$

Given that

$$\overrightarrow{OA} = \begin{pmatrix} 5 \\ -3 \\ -4 \end{pmatrix} \quad \text{and} \quad \mathbf{n} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

where O is the origin,

- (a) find a Cartesian equation of Π .

(2)

With respect to the fixed origin O , the line l is given by the equation

$$\mathbf{r} = \begin{pmatrix} 7 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -5 \\ 3 \end{pmatrix}$$

The line l intersects the plane Π at the point X .

- (b) Show that the acute angle between the plane Π and the line l is 21.2° correct to one decimal place.

(4)

- (c) Find the coordinates of the point X .

(4)

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Question 2 continued

(Total for Question 2 is 10 marks)

3. Tyler invested a total of £5000 across three different accounts; a savings account, a property bond account and a share dealing account.

Tyler invested £400 more in the property bond account than in the savings account.

After one year

- the savings account had increased in value by 1.5%
- the property bond account had increased in value by 3.5%
- the share dealing account had **decreased** in value by 2.5%
- the total value across Tyler's three accounts had increased by £79

Form and solve a matrix equation to find out how much money was invested by Tyler in each account.

(7)

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4. The cubic equation

$$x^3 + 3x^2 - 8x + 6 = 0$$

has roots α , β and γ .

Without solving the equation, find the cubic equation whose roots are $(\alpha - 1)$, $(\beta - 1)$ and $(\gamma - 1)$, giving your answer in the form $w^3 + pw^2 + qw + r = 0$, where p , q and r are integers to be found.

(5)

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5.

$$\mathbf{M} = \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$$

- (a) Show that $\mathbf{M}$ is non-singular.

(2)

The hexagon R is transformed to the hexagon S by the transformation represented by the matrix $\mathbf{M}$.

Given that the area of hexagon R is 5 square units,

- (b) find the area of hexagon S .

(1)

The matrix $\mathbf{M}$ represents an enlargement, with centre $(0, 0)$ and scale factor k , where $k > 0$, followed by a rotation anti-clockwise through an angle θ about $(0, 0)$.

- (c) Find the value of k .

(2)

- (d) Find the value of θ .

(2)

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Question 6 continued

(Total for Question 6 is 15 marks)

7.

Diagrams not drawn to scale

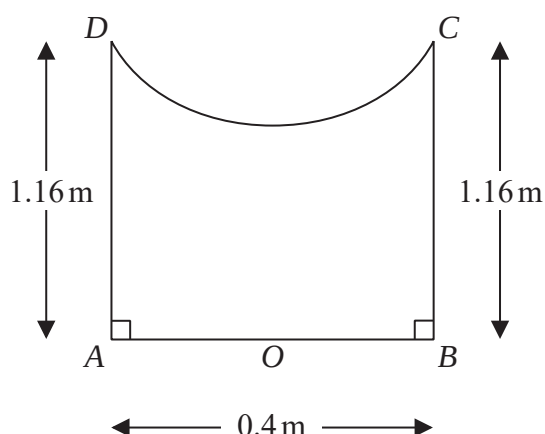


Figure 1

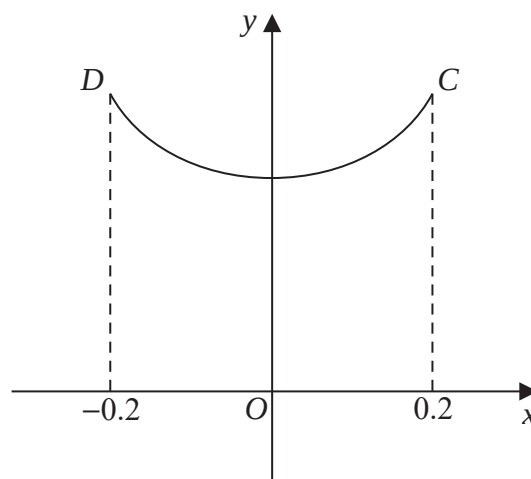


Figure 2

Figure 1 shows the central cross-section $AOBCD$ of a circular bird bath, which is made of concrete. Measurements of the height and diameter of the bird bath, and the depth of the bowl of the bird bath have been taken in order to estimate the amount of concrete that was required to make this bird bath.

Using these measurements, the cross-sectional curve CD , shown in Figure 2, is modelled as a curve with equation

$$y = 1 + kx^2 \quad -0.2 \leq x \leq 0.2$$

where k is a constant and where O is the fixed origin.

The height of the bird bath measured 1.16 m and the diameter, AB , of the base of the bird bath measured 0.40 m, as shown in Figure 1.

- Suggest the maximum depth of the bird bath. (1)
- Find the value of k . (2)
- Hence find the volume of concrete that was required to make the bird bath according to this model. Give your answer, in m^3 , correct to 3 significant figures. (7)
- State a limitation of the model. (1)

It was later discovered that the volume of concrete used to make the bird bath was 0.127 m^3 correct to 3 significant figures.

- Using this information and the answer to part (c), evaluate the model, explaining your reasoning. (1)

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8. (a) Shade on an Argand diagram the set of points

$$\left\{ z \in \mathbb{C} : |z - 4i| \leq 3 \right\} \cap \left\{ z \in \mathbb{C} : -\frac{\pi}{2} < \arg(z + 3 - 4i) \leq \frac{\pi}{4} \right\}$$

(6)

The complex number w satisfies

$$|w - 4i| = 3$$

- (b) Find the maximum value of $\arg w$ in the interval $(-\pi, \pi]$.
Give your answer in radians correct to 2 decimal places.

(2)

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9. An octopus is able to catch any fish that swim within a distance of 2 m from the octopus's position.

A fish F swims from a point A to a point B .

The octopus is modelled as a fixed particle at the origin O .

Fish F is modelled as a particle moving in a straight line from A to B .

Relative to O , the coordinates of A are $(-3, 1, -7)$ and the coordinates of B are $(9, 4, 11)$, where the unit of distance is metres.

- (a) Use the model to determine whether or not the octopus is able to catch fish F . (7)
- (b) Criticise the model in relation to fish F . (1)
- (c) Criticise the model in relation to the octopus. (1)

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Please check the examination details below before entering your candidate information

Candidate surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Specimen Paper

(Time: 1 hour 40 minutes)

Paper Reference **8FM0/01**

Further Mathematics

Advanced Subsidiary

Paper 1: Core Pure Mathematics

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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(3)

Question 1 continued

(Total for Question 1 is 5 marks)



2. (a) Sketch, on an Argand diagram, the set of points

$$X = \{z \in \mathbb{C} : |z - 4 - 2i| < 3\} \cap \left\{z \in \mathbb{C} : 0 \leq \arg(z) \leq \frac{\pi}{4}\right\}$$

On your diagram

- shade the part of the diagram that is included in the set
- use solid lines to show the parts of the boundary that are included in the set, and use dashed lines to show the parts of the boundary that are not included in the set

(3)

(b) Show that the complex number $z = 5 + 4i$ is in the set X .

(3)



Question 2 continued

(Total for Question 2 is 6 marks)



3. (a) Find, in terms of the real constant k , the determinant of the matrix

$$\mathbf{M} = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 3 & -1 \\ 1 & k & 2 \end{pmatrix} \quad (2)$$

Three distinct planes, Π_1 , Π_2 and Π_3 , are defined by the equations

$$\begin{aligned} \Pi_1 &: \mathbf{r} \cdot \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = 4 \\ \Pi_2 &: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \\ \Pi_3 &: x + ky + 2z = -1 \end{aligned}$$

where λ and μ are scalar parameters.

(b) Find an equation in Cartesian form for

- $$\begin{aligned} \text{(i)} \quad & \Pi_1 \\ \text{(ii)} \quad & \Pi_2 \end{aligned} \tag{4}$$

Given that the three planes Π_1 , Π_2 and Π_3 form a sheaf,

(c) use the answer to part (a) to explain why $k = -1$ (2)



Question 3 continued

Handwriting practice area with 20 horizontal lines.

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Question 3 continued

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Question 3 continued

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(Total for Question 3 is 8 marks)



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4. A company manufacturing radios agreed a 20 year contract with a retailer to supply its radios. In the first year of the contract, 500 radios were supplied to the retailer. In each subsequent year, the number of radios supplied to the retailer was 50 more than in the previous year.

The amount received by the company for each radio during year n of the contract was £ $\left(20 + \frac{n^2}{45}\right)$

The total cost of producing the radios during year n was modelled as $\pounds(1000 + 10n^2)$

- (a) Show that, according to the model, the profit made by the company in year n , $\text{£}P_n$, is given by

$$P_n = \frac{10}{9}(n^3 + 900n + 7200) \quad (2)$$

- (b) Use the standard results for summations to show that the total profit made by the company in the first N years of the contract, $\text{£}T_N$, is given by

$$T_N = aN(N^3 + bN^2 + cN + d)$$

where a, b, c and d are constants to be found.

At the end of the 20 years, the company found that its total profit made from this contract just exceeded £500 000.

- (c) Assess the model in light of this information.



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Question 4 continued

Lined area for writing the answer to Question 4.

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Question 4 continued

(Total for Question 4 is 9 marks)



5.

$$f(z) = 8z^3 + 12z^2 + 6z + 65$$

Given that $\frac{1}{2} - i\sqrt{3}$ is a root of the equation $f(z) = 0$

- (a) write down the other complex root of the equation, (1)
- (b) use algebra to solve the equation $f(z) = 0$ completely. (3)
- (c) Show the roots of $f(z)$ on a single Argand diagram. (2)
- (d) Show that the roots of $f(z)$ form the vertices of an equilateral triangle in the complex plane. (2)

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Question 5 continued

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Question 5 continued

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Question 5 continued

(Total for Question 5 is 8 marks)



6. (a) Prove by induction that, for all $n \in \mathbb{Z}^+$

$$f(n) = n^5 + 4n$$

is divisible by 5

(6)

(b) Show that $f(-x) = -f(x)$ for all $x \in \mathbb{R}$

(1)

(c) Hence prove that $f(n)$ is divisible by 5 for all $n \in \mathbb{Z}$

(2)

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Question 6 continued

(Total for Question 6 is 9 marks)



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7. The population of Zebu cattle in a particular country is modelled by two sub-populations, adults and juveniles. In this model, the only factors affecting the population of the Zebu are the birth and survival rates of the population.

Data recorded in the years preceding 2018 was used to suggest the annual birth and survival rates of the population.

The results are shown in the table below, with values to 2 significant figures. It is assumed that these rates will remain the same in future years.

	Birth rate	Survival rate
Adult population	0.23	0.97
Juvenile population	0	0.87

It is also assumed that $\frac{1}{3}$ of the surviving juvenile population become adults each year.

Let A_n and J_n be the respective sub-populations, in millions, of adults and juveniles, n years after 1st January 2018. Then the adult population in year $n + 1$ satisfies the equation

$$A_{n+1} = 0.97A_n + \frac{1}{3}(0.87)J_n = 0.97A_n + 0.29J_n$$

- (a) Form the corresponding equation for the juvenile population in year $n + 1$ under this model, justifying your values.

(2)

The total population on 1st January 2018 was estimated, to 2 significant figures, as 1.5 million Zebu, with 1.2 million of these being adults.

- (b) Find the value of p and the matrix $\mathbf{M}$ such that the population of Zebu can be modelled by the system

$$\begin{pmatrix} A_0 \\ J_0 \end{pmatrix} = \begin{pmatrix} 1.2 \\ p \end{pmatrix} \quad \begin{pmatrix} A_{n+1} \\ J_{n+1} \end{pmatrix} = \mathbf{M} \begin{pmatrix} A_n \\ J_n \end{pmatrix}$$

Give p to 2 significant figures and each entry of $\mathbf{M}$ to 2 decimal places.

(3)

Using the model formed in (b), find, to 3 significant figures,

- (c) (i) the **total** Zebu population that was present on 1st January 2017
(ii) the predicted **juvenile** Zebu population on 1st January 2025

(5)

As a result of the predictions of this model, it is decided that the country will export 15 000 juveniles to a neighbouring country at the end of each year.

- (d) Adapt the model from 2018 onwards to include this export.

(2)

- (e) State one limitation of this model.

(1)



Turn over ►



Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

(Total for Question 7 is 13 marks)



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Question 8 continued

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Question 8 continued

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Question 8 continued

(Total for Question 8 is 11 marks)



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9. A small comet C is passing near to a planet. The planet can be modelled as a sphere with centre O taken as a fixed point in space, so that the motion of the comet is relative to the origin O .

The diameter of the planet is 13 000 km.

The comet is monitored by satellites orbiting the planet.

When the monitoring begins the comet is at position $146\mathbf{i} + 234\mathbf{j} - 85\mathbf{k}$ and is moving with vector $-21\mathbf{i} - 33\mathbf{j} + 13\mathbf{k}$ every hour, where the units are in thousands of kilometres.

Assuming the comet maintains a straight line course throughout its motion,

- (a) determine whether or not the comet will collide with the planet. (6)

Two of the satellites, A and B , have position vectors $\vec{OA} = 5\mathbf{i} + 12\mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 12\mathbf{j} - 3\mathbf{k}$ at the beginning of monitoring. They return to these positions every 4 hours.

- (b) Find the expected angle ACB between the comet and the satellites A and B when they first return to their initial positions. Give your answer to the nearest 0.1°

- (c) Give a reason why the answer to (b) may differ from the true value. (1)



Question 9 continued

Handwriting practice area with 20 horizontal lines.

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Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Handwriting practice area with 20 horizontal lines.



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Question 9 continued

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(Total for Question 9 is 11 marks)

TOTAL FOR PAPER IS 80 MARKS

